

Groups and Lie Algebras: Solutions to Problem Sheet 9

1 Gell–Mann matrices of $\mathfrak{sl}(3)$

Problem 1. Let $E_{ij} \in \mathfrak{gl}(3, \mathbb{C})$ be the matrix unit with entries

$$(E_{ij})_{kl} = \delta_{ik}\delta_{jl}.$$

(a) Show that

$$E_{ij}E_{kl} = \delta_{jk}E_{il},$$

and deduce

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}.$$

(b) Show that the six off-diagonal units E_{ij} , $i \neq j$, together with

$$H_{12} = E_{11} - E_{22}, \quad H_{23} = E_{22} - E_{33},$$

form a basis of $\mathfrak{sl}(3, \mathbb{C})$. Express the Gell–Mann ladder combinations

$$\frac{1}{2}(\lambda_1 \pm i\lambda_2), \quad \frac{1}{2}(\lambda_4 \pm i\lambda_5), \quad \frac{1}{2}(\lambda_6 \pm i\lambda_7)$$

in terms of the E_{ij} .

Matrix units and their product

The matrix E_{ij} has a single nonzero entry: a 1 in row i , column j . To compute the product $E_{ij}E_{kl}$, we compute its (a, b) -entry:

$$(E_{ij}E_{kl})_{ab} = \sum_c (E_{ij})_{ac}(E_{kl})_{cb}.$$

Using

$$(E_{ij})_{ac} = \delta_{ia}\delta_{jc}, \quad (E_{kl})_{cb} = \delta_{kc}\delta_{lb},$$

we get

$$\begin{aligned} (E_{ij}E_{kl})_{ab} &= \sum_c \delta_{ia}\delta_{jc}\delta_{kc}\delta_{lb} \\ &= \delta_{ia}\delta_{lb} \sum_c \delta_{jc}\delta_{kc} \\ &= \delta_{ia}\delta_{lb}\delta_{jk}. \end{aligned}$$

If $j \neq k$, this is zero. If $j = k$, it is exactly the (a, b) -entry of E_{il} . Hence

$$\boxed{E_{ij}E_{kl} = \delta_{jk}E_{il}.}$$

The commutator in $\mathfrak{gl}(3, \mathbb{C})$ is defined by

$$[X, Y] = XY - YX.$$

Therefore

$$\begin{aligned} [E_{ij}, E_{kl}] &= E_{ij}E_{kl} - E_{kl}E_{ij} \\ &= \delta_{jk}E_{il} - \delta_{li}E_{kj}. \end{aligned}$$

Thus

$$\boxed{[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}.}$$

A basis of $\mathfrak{sl}(3, \mathbb{C})$

Recall that

$$\mathfrak{sl}(3, \mathbb{C}) = \{X \in \text{Mat}_3(\mathbb{C}) : \text{tr } X = 0\}.$$

The full vector space $\text{Mat}_3(\mathbb{C})$ has basis

$$\{E_{ij} : 1 \leq i, j \leq 3\},$$

so it has dimension 9. The trace condition imposes one independent linear condition, namely

$$X_{11} + X_{22} + X_{33} = 0.$$

Hence

$$\dim \mathfrak{sl}(3, \mathbb{C}) = 8.$$

The six off-diagonal matrices are

$$E_{12}, E_{21}, E_{13}, E_{31}, E_{23}, E_{32}.$$

These account for the off-diagonal entries of a traceless matrix. The diagonal part of a traceless matrix has the form

$$\text{diag}(a, b, c), \quad a + b + c = 0.$$

This is a two-dimensional vector space. A convenient basis is

$$H_{12} = E_{11} - E_{22} = \text{diag}(1, -1, 0),$$

$$H_{23} = E_{22} - E_{33} = \text{diag}(0, 1, -1).$$

Indeed, any traceless diagonal matrix can be written as a linear combination of these. If

$$xH_{12} + yH_{23} = \text{diag}(x, -x + y, -y),$$

then to match $\text{diag}(a, b, c)$ we need

$$x = a, \quad -y = c, \quad -x + y = b.$$

The last equation becomes $-a - c = b$, which is true because $a + b + c = 0$. Thus

$$\text{diag}(a, b, c) = aH_{12} - cH_{23}.$$

Therefore the six off-diagonal units together with H_{12}, H_{23} form a basis of $\mathfrak{sl}(3, \mathbb{C})$.

Gell–Mann ladder combinations

The first two Gell–Mann matrices are

$$\lambda_1 = E_{12} + E_{21}, \quad \lambda_2 = -iE_{12} + iE_{21}.$$

Therefore

$$\begin{aligned} \frac{1}{2}(\lambda_1 + i\lambda_2) &= \frac{1}{2}(E_{12} + E_{21} + i(-iE_{12} + iE_{21})) \\ &= \frac{1}{2}(E_{12} + E_{21} + E_{12} - E_{21}) \\ &= E_{12}, \end{aligned}$$

and similarly

$$\frac{1}{2}(\lambda_1 - i\lambda_2) = E_{21}.$$

In the same way,

$$\lambda_4 = E_{13} + E_{31}, \quad \lambda_5 = -iE_{13} + iE_{31},$$

so

$$\boxed{\frac{1}{2}(\lambda_4 + i\lambda_5) = E_{13}, \quad \frac{1}{2}(\lambda_4 - i\lambda_5) = E_{31}.}$$

Finally,

$$\lambda_6 = E_{23} + E_{32}, \quad \lambda_7 = -iE_{23} + iE_{32},$$

and hence

$$\boxed{\frac{1}{2}(\lambda_6 + i\lambda_7) = E_{23}, \quad \frac{1}{2}(\lambda_6 - i\lambda_7) = E_{32}.}$$

Thus the Gell–Mann matrices are Hermitian combinations of the root vectors E_{ij} , while the ladder combinations recover the standard root-vector basis of $\mathfrak{sl}(3, \mathbb{C})$.

2 Tensor products of $\mathfrak{sl}(3)$ -representations

Problem 2. Decompose the tensor product of the 8-dimensional adjoint representation of $\mathfrak{sl}(3)$ with the defining three-dimensional representation on $\mathbb{C}^3$ into irreducible representations.

The defining representation is denoted by **3**. In Dynkin-label notation for $\mathfrak{sl}(3)$, it is

$$\mathbf{3} = (1, 0).$$

The dual defining representation is

$$\bar{\mathbf{3}} = (0, 1),$$

and the adjoint representation is

$$\mathbf{8} = (1, 1).$$

We want to decompose

$$\mathbf{8} \otimes \mathbf{3}.$$

A useful way to compute this is to use the standard identity

$$\mathbf{3} \otimes \bar{\mathbf{3}} \cong \mathbf{8} \oplus \mathbf{1}.$$

This is the statement that

$$\text{End}(\mathbb{C}^3) \cong \mathfrak{sl}(3, \mathbb{C}) \oplus \mathbb{C} \text{id}.$$

Indeed, every endomorphism of $\mathbb{C}^3$ decomposes uniquely into its traceless part plus its scalar part.

Now tensor both sides with $\mathbf{3}$:

$$(\mathbf{3} \otimes \bar{\mathbf{3}}) \otimes \mathbf{3} \cong (\mathbf{8} \oplus \mathbf{1}) \otimes \mathbf{3} \cong (\mathbf{8} \otimes \mathbf{3}) \oplus \mathbf{3}.$$

We compute the left-hand side in another order:

$$(\mathbf{3} \otimes \mathbf{3}) \otimes \bar{\mathbf{3}}.$$

The tensor square of the defining representation decomposes into symmetric and antisymmetric parts:

$$\mathbf{3} \otimes \mathbf{3} \cong \mathbf{6} \oplus \bar{\mathbf{3}}.$$

Here $\mathbf{6} = (2, 0)$ is the symmetric square $\text{Sym}^2 \mathbb{C}^3$, while $\bar{\mathbf{3}} = (0, 1)$ is $\Lambda^2 \mathbb{C}^3$, which is naturally dual to $\mathbb{C}^3$ for $\mathfrak{sl}(3)$.

Thus

$$(\mathbf{3} \otimes \mathbf{3}) \otimes \bar{\mathbf{3}} \cong (\mathbf{6} \oplus \bar{\mathbf{3}}) \otimes \bar{\mathbf{3}} \cong (\mathbf{6} \otimes \bar{\mathbf{3}}) \oplus (\bar{\mathbf{3}} \otimes \bar{\mathbf{3}}).$$

The standard $\mathfrak{sl}(3)$ tensor product rules give

$$\mathbf{6} \otimes \bar{\mathbf{3}} \cong \mathbf{15} \oplus \mathbf{3},$$

and

$$\bar{\mathbf{3}} \otimes \bar{\mathbf{3}} \cong \bar{\mathbf{6}} \oplus \mathbf{3}.$$

Therefore

$$(\mathbf{3} \otimes \bar{\mathbf{3}}) \otimes \mathbf{3} \cong \mathbf{15} \oplus \mathbf{3} \oplus \bar{\mathbf{6}} \oplus \mathbf{3}.$$

On the other hand we already found

$$(\mathbf{3} \otimes \bar{\mathbf{3}}) \otimes \mathbf{3} \cong (\mathbf{8} \otimes \mathbf{3}) \oplus \mathbf{3}.$$

Cancelling the common $\mathbf{3}$ summand gives

$$\boxed{\mathbf{8} \otimes \mathbf{3} \cong \mathbf{15} \oplus \bar{\mathbf{6}} \oplus \mathbf{3}.$$

In Dynkin labels, this reads

$$(1, 1) \otimes (1, 0) \cong (2, 1) \oplus (0, 2) \oplus (1, 0).$$

As a dimension check, recall that the irreducible $\mathfrak{sl}(3)$ -representation with highest weight (p, q) has dimension

$$\dim(p, q) = \frac{(p+1)(q+1)(p+q+2)}{2}.$$

Thus

$$\dim(2, 1) = \frac{3 \cdot 2 \cdot 5}{2} = 15,$$

$$\dim(0, 2) = \frac{1 \cdot 3 \cdot 4}{2} = 6,$$

$$\dim(1, 0) = \frac{2 \cdot 1 \cdot 3}{2} = 3.$$

The right-hand side has dimension

$$15 + 6 + 3 = 24,$$

which agrees with

$$\dim(\mathbf{8} \otimes \mathbf{3}) = 8 \cdot 3 = 24.$$

3 The Weyl group of $\mathfrak{sl}(3)$ and invariance of weight diagrams

Problem 3. Let $\alpha_1 = L_1 - L_2$ and $\alpha_2 = L_2 - L_3$ be the simple roots of $\mathfrak{sl}(3, \mathbb{C})$. The simple reflections act by

$$s_1(\alpha_1) = -\alpha_1, \quad s_1(\alpha_2) = \alpha_1 + \alpha_2,$$

and

$$s_2(\alpha_1) = \alpha_1 + \alpha_2, \quad s_2(\alpha_2) = -\alpha_2.$$

- (a) Draw the action of the Weyl group on the root lattice, verify $s_1^2 = s_2^2 = e$ and $(s_1 s_2)^3 = e$, and conclude that $W \cong S_3$.
- (b) Use the representation theory of $\mathfrak{sl}(2, \mathbb{C})$ to show that weight multiplicities are invariant under the Weyl group.
- (c) Check this for the adjoint representation and for the fundamental representation.
- (d) Is there a larger symmetry group which acts on all weight diagrams?

The simple reflections as matrices

Write a general element of the root lattice as

$$\mu = a\alpha_1 + b\alpha_2.$$

The reflections are given by

$$s_1(\mu) = (b - a)\alpha_1 + b\alpha_2,$$

$$s_2(\mu) = a\alpha_1 + (a - b)\alpha_2.$$

In the ordered basis (α_1, α_2) , the matrices are therefore

$$s_1 = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}.$$

Indeed, the first column of the matrix of s_1 is the coordinate vector of $s_1(\alpha_1) = -\alpha_1$, and the second column is the coordinate vector of $s_1(\alpha_2) = \alpha_1 + \alpha_2$. Similarly for s_2 .

Now compute

$$s_1^2 = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and

$$s_2^2 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Next,

$$s_1 s_2 = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

One checks that

$$(s_1 s_2)^2 = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence

$$(s_1 s_2)^3 = I.$$

Thus

$$s_1^2 = s_2^2 = e, \quad (s_1 s_2)^3 = e.$$

These are the Coxeter relations of type A_2 . The group generated by two reflections satisfying these relations is the symmetry group of an equilateral triangle, hence is isomorphic to S_3 . Therefore

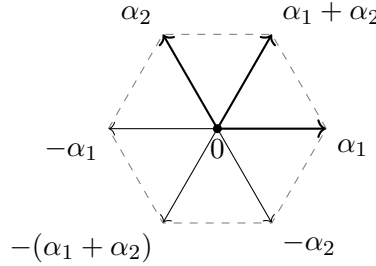
$$W(\mathfrak{sl}(3)) \cong S_3.$$

The root lattice picture

The nonzero roots of $\mathfrak{sl}(3)$ are

$$\pm\alpha_1, \quad \pm\alpha_2, \quad \pm(\alpha_1 + \alpha_2).$$

They form a hexagon. A schematic picture is:



The Weyl group acts by reflecting this hexagon in the hyperplanes perpendicular to the simple roots. It permutes the six nonzero roots transitively.

Weyl invariance of weight multiplicities

Let V be a finite-dimensional $\mathfrak{sl}(3, \mathbb{C})$ -representation with weight space decomposition

$$V = \bigoplus_{\mu} V_{\mu},$$

where

$$V_{\mu} = \{v \in V : Hv = \mu(H)v \text{ for all } H \in \mathfrak{h}\}.$$

We want to show that

$$\dim V_{\mu} = \dim V_{s_i \mu}$$

for the simple reflections s_1, s_2 , and therefore for every element of the Weyl group.

Fix a simple root α , for example α_1 . Associated to it is an $\mathfrak{sl}(2, \mathbb{C})$ -subalgebra

$$\langle E_{\alpha}, E_{-\alpha}, H_{\alpha} \rangle.$$

For α_1 , this is

$$\langle E_{12}, E_{21}, H_{12} \rangle,$$

and for α_2 , this is

$$\langle E_{23}, E_{32}, H_{23} \rangle.$$

Now restrict the representation V from $\mathfrak{sl}(3, \mathbb{C})$ to this $\mathfrak{sl}(2, \mathbb{C})$ -subalgebra. Since finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$ -modules are completely reducible, V decomposes into a direct sum of irreducible $\mathfrak{sl}(2, \mathbb{C})$ -modules.

Inside an irreducible $\mathfrak{sl}(2, \mathbb{C})$ -module, the weights are symmetric under the reflection

$$m \mapsto -m.$$

Indeed, the irreducible representation of highest weight N has weights

$$N, N-2, N-4, \dots, -N.$$

Each of these occurs with multiplicity one, and the list is symmetric about zero.

For the $\mathfrak{sl}(3)$ -weight μ , the corresponding $\mathfrak{sl}(2)$ -weight is

$$\mu(H_\alpha).$$

The simple reflection acts by reflecting this component:

$$s_\alpha(\mu) = \mu - \mu(H_\alpha)\alpha.$$

Thus, along each α -string of weights, s_α reflects the string in the same way that $m \mapsto -m$ reflects the weights of an $\mathfrak{sl}(2)$ -representation.

Because $\mathfrak{sl}(2)$ -weight multiplicities are symmetric, we obtain

$$\dim V_\mu = \dim V_{s_\alpha(\mu)}.$$

Applying this to α_1 and α_2 gives invariance under s_1 and s_2 . Since the Weyl group is generated by s_1, s_2 , the set of weights of V , counted with multiplicity, is invariant under all of W .

Check: the adjoint representation

For the adjoint representation

$$V = \mathfrak{sl}(3, \mathbb{C}),$$

the weights are the roots, together with the zero weight coming from the Cartan subalgebra:

$$\text{wt}(\mathfrak{sl}(3, \mathbb{C})) = \{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2), 0, 0\}.$$

The zero weight has multiplicity 2, because

$$\mathfrak{h} = \text{span}(H_{12}, H_{23})$$

is two-dimensional. The six nonzero roots form one Weyl orbit, and the zero weight is fixed by the Weyl group. Thus the adjoint weight diagram is Weyl invariant.

Check: the fundamental representation

For the defining representation

$$V = \mathbb{C}^3,$$

with basis vectors e_1, e_2, e_3 , the weights are

$$L_1, L_2, L_3,$$

where L_i evaluates a diagonal matrix on its i -th diagonal entry. For example, if

$$H = \text{diag}(h_1, h_2, h_3), \quad h_1 + h_2 + h_3 = 0,$$

then

$$He_i = h_i e_i,$$

so e_i has weight L_i .

The Weyl group of $\mathfrak{sl}(3)$ permutes the three coordinate functionals

$$L_1, L_2, L_3.$$

Thus the three weights of the fundamental representation form a single Weyl orbit. In particular, the weight diagram of $\mathbb{C}^3$ is invariant under $W \cong S_3$.

Is there a larger symmetry?

The Weyl group is the symmetry group that acts on the weights of every finite-dimensional representation, preserving multiplicities.

For $\mathfrak{sl}(3)$, the full symmetry group of the root hexagon is larger: it is the dihedral group of order 12. This includes the Weyl group of order 6 and also the diagram automorphism exchanging the two simple roots

$$\alpha_1 \leftrightarrow \alpha_2.$$

However, this extra symmetry does not preserve the weight diagram of every single representation. It sends a representation with highest weight (p, q) to the dual representation with highest weight (q, p) . Therefore it is a symmetry of a particular weight diagram only when the representation is self-dual, for instance when $p = q$, or in special cases such as the adjoint representation.

Thus the universal symmetry of all weight diagrams is the Weyl group. A larger symmetry exists for the root system, but not for every individual representation.

4 Sorting hadrons into $SU(3)$ multiplets

Problem 4. The sheet gives a list of seventeen hadrons with electric charge Q , strangeness S , baryon number B , and approximate mass. Compute

$$Y = B + S, \quad I_3 = Q - \frac{1}{2}Y,$$

and use these values to organise the particles into $SU(3)$ -multiplets.

Why mass alone is not enough

If one tries to organise the particles only by mass, some rough clusters appear, for example the pions near 140 MeV, the kaons near 500 MeV, the nucleons near 940 MeV, and the Σ -baryons near 1190 MeV. However, this does not reveal a clean $SU(3)$ -multiplet structure.

The reason is that mass is not itself a weight coordinate of the $SU(3)$ flavour representation. The weight diagram is described instead by

$$(I_3, Y),$$

where I_3 is the third component of isospin and Y is hypercharge. The masses are only approximately related within a multiplet because flavour $SU(3)$ is not an exact symmetry in nature. It is broken by the different quark masses, especially the strange quark mass.

Computing Y and I_3

For each particle we compute

$$Y = B + S, \quad I_3 = Q - \frac{1}{2}Y.$$

The results are as follows. The entry denoted $\bar{K}^0$ is the neutral anti-kaon; in the sheet it appears as the neutral kaon with $S = -1$.

Particle	Q	S	B	Mass	$Y = B + S$	$I_3 = Q - \frac{1}{2}Y$
Σ^0	0	-1	1	1193	0	0
K^+	1	1	0	494	1	$\frac{1}{2}$
n	0	0	1	940	1	$-\frac{1}{2}$
π^-	-1	0	0	140	0	-1
Ξ^-	-1	-2	1	1322	-1	$-\frac{1}{2}$
η	0	0	0	548	0	0
Σ^+	1	-1	1	1189	0	1
K^0	0	1	0	498	1	$-\frac{1}{2}$
Λ	0	-1	1	1116	0	0
π^+	1	0	0	140	0	1
Ξ^0	0	-2	1	1315	-1	$\frac{1}{2}$
η'	0	0	0	958	0	0
p	1	0	1	938	1	$\frac{1}{2}$
K^-	-1	-1	0	494	-1	$-\frac{1}{2}$
Σ^-	-1	-1	1	1197	0	-1
π^0	0	0	0	135	0	0
$\bar{K}^0$	0	-1	0	498	-1	$\frac{1}{2}$

If all seventeen particles are plotted at once on the (I_3, Y) -plane, many of them overlap. This is because (I_3, Y) does not remember the baryon number B separately. For example, both mesons and baryons may have the same values of (I_3, Y) , but they belong to different physical sectors.

Thus the column which separates the particles before plotting the weight diagrams is

$B, \text{ the baryon number.}$

The particles with $B = 1$ are baryons, while the particles with $B = 0$ are mesons.

The baryon octet

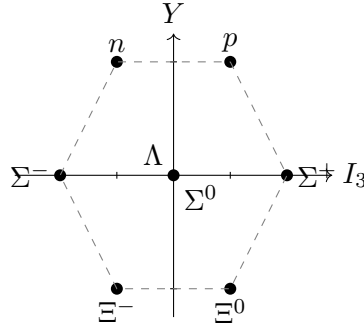
The baryons are

$$p, n, \Sigma^+, \Sigma^0, \Sigma^-, \Lambda, \Xi^0, \Xi^-.$$

Their (I_3, Y) -coordinates are:

Particle	(I_3, Y)
p	$(\frac{1}{2}, 1)$
n	$(-\frac{1}{2}, 1)$
Σ^+	$(1, 0)$
Σ^0	$(0, 0)$
Σ^-	$(-1, 0)$
Λ	$(0, 0)$
Ξ^0	$(\frac{1}{2}, -1)$
Ξ^-	$(-\frac{1}{2}, -1)$

This is the familiar baryon octet. The zero weight $(0,0)$ occurs with multiplicity 2, represented physically by Σ^0 and Λ .



The baryon octet in the (I_3, Y) -plane.

The meson octet plus singlet

The mesons are

$$K^+, K^0, K^-, \bar{K}^0, \pi^+, \pi^0, \pi^-, \eta, \eta'.$$

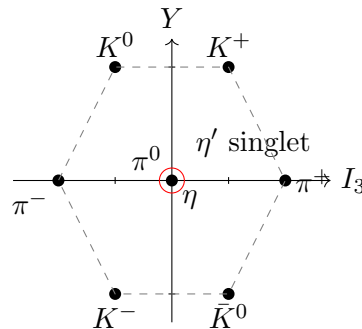
Their (I_3, Y) -coordinates are:

Particle	(I_3, Y)
K^+	$(\frac{1}{2}, 1)$
K^0	$(-\frac{1}{2}, 1)$
π^+	$(1, 0)$
π^0	$(0, 0)$
η	$(0, 0)$
η'	$(0, 0)$
π^-	$(-1, 0)$
$\bar{K}^0$	$(\frac{1}{2}, -1)$
K^-	$(-\frac{1}{2}, -1)$

Here three particles sit at the origin:

$$\pi^0, \quad \eta, \quad \eta'.$$

But the adjoint representation $\mathbf{8}$ of $SU(3)$ has zero weight multiplicity 2. Therefore only two independent particles at the origin can belong to the octet. The third must be an $SU(3)$ -singlet.



The meson octet, together with an additional singlet at the origin.

Why the singlet sits at the origin

The mesons can be modelled as quark–antiquark states. In $SU(3)$ flavour language this gives

$$\mathbf{3} \otimes \bar{\mathbf{3}}.$$

As above,

$$\mathbf{3} \otimes \bar{\mathbf{3}} \cong \mathbf{8} \oplus \mathbf{1}.$$

The singlet representation $\mathbf{1}$ is one-dimensional and trivial. Since all elements of the Cartan subalgebra act by zero on the trivial representation, its weight is necessarily

$$(I_3, Y) = (0, 0).$$

Thus the singlet is forced to sit at the origin of the weight diagram.

Among the three neutral mesons at the origin,

$$\pi^0, \quad \eta, \quad \eta',$$

the two octet states are naturally identified with π^0 and η , while the extra singlet is naturally identified with η' . This assignment is also physically reasonable because η' is substantially heavier, reflecting the fact that the physical η and η' are mixtures of the ideal octet and singlet states.

Therefore the most natural assignment is

η' belongs to the $SU(3)$ singlet $\mathbf{1}$.
